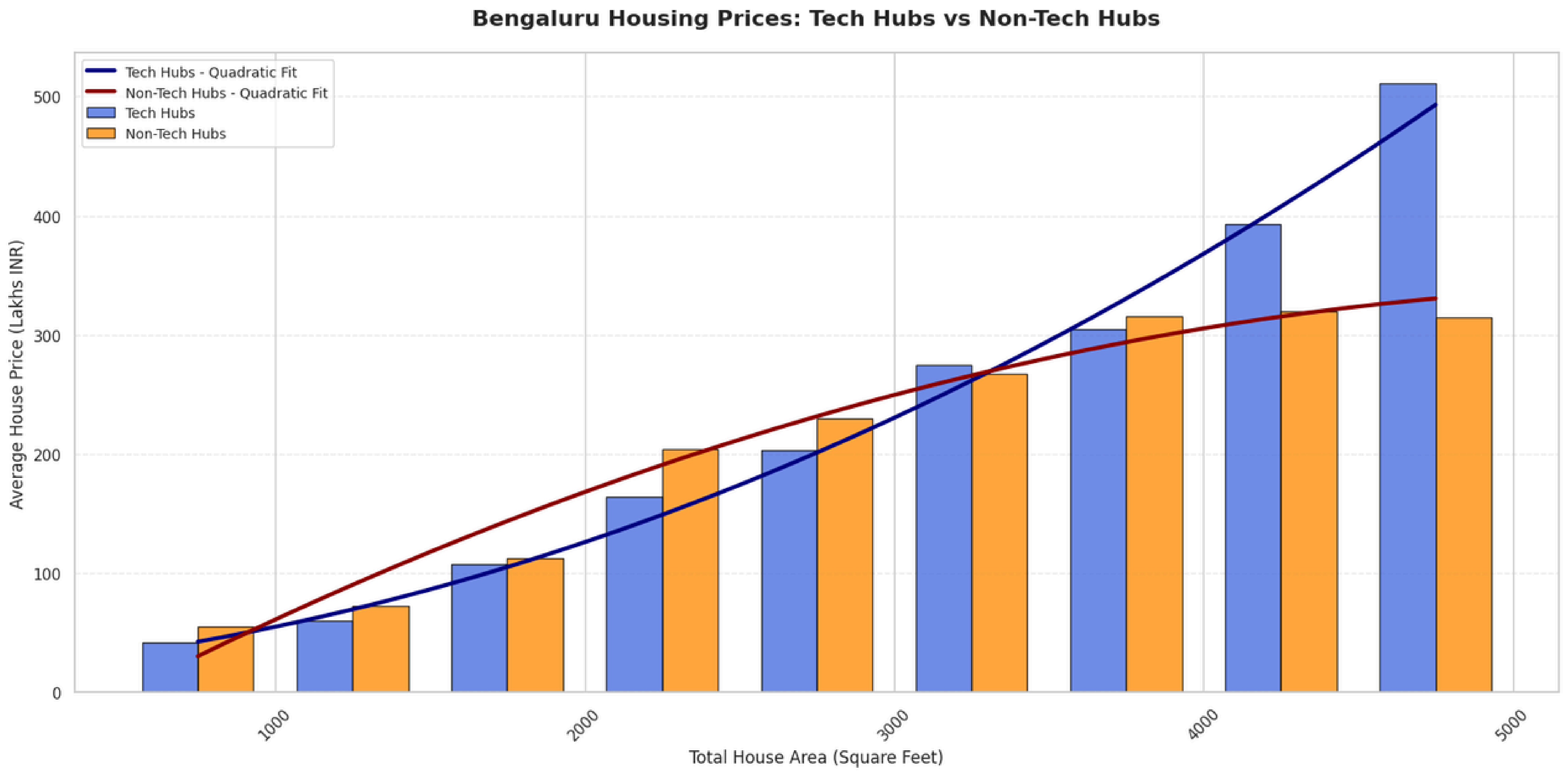
MA26M003
MA26M018

Bengaluru Housing Prices: Tech Hubs vs Other Locations

Statistical investigation of residential property prices in Bengaluru



- **Methods:** Welch’s Two-Sample t-Test and Mann–Whitney U Test
- **Key result:** Both tests reveal statistically significant differences between the two groups.
- **Takeaway:** Housing prices exhibit different distributional behaviour across Tech Hub and other locations.

Data Source: https://www.kaggle.com/datasets/sumanhachappa/banglore-house-rent

Richard Price
(1723-1791)

Thomas Bayes
(1702-1761)

The Bayesian Analysis of the Resurrection of Jesus Christ :
The Origin of Bayes Theorem

David Hume
(1711- 1776)

Mr Presenter
(2003 -)

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01. Historical Context

- **Thomas Bayes** : Developed the idea of updating probabilities.
- **Richard Price** : Edited and published Bayes' work, linking it to the debate on miracles.
- **David Hume** : Argued that the probability of a miracle is so low that testimony can never establish it.

02. Evidences

- **Martyrdom** : Followers were willing to die for their claim.
- **Empty Tomb** : Reports and witnesses of the empty tomb and appearances.
- **Early Writings** : Letters and texts written close to the events.

03. Bayesian Framework

$$\underbrace{\frac{P(H\mid E,B)}{P(\neg H\mid E,B)}}_{\text{Posterior odds}} = \underbrace{\frac{P(H\mid B)}{P(\neg H\mid B)}}_{\text{Prior odds}} \times \underbrace{\frac{P(E\mid H,B)}{P(E\mid \neg H,B)}}_{\text{likelihood}}$$

Decomposition of the Evidence (Chain Rule) :

For multiple evidence components: $E = E_1 \cap E_2 \cap \ldots E_n$

$$P(E\mid H,B) = \prod_{k=1}^{n} P(E_k \mid E_1,\ldots,E_{k-1},H,B)$$

Here, we account for dependence between evidence

Decomposition of Alternative Hypotheses :

Let A_i s be mutually exclusive and collectively exhaustive alternatives:

$\neg H = A_1 \cup A_2 \ldots \cup A_n$

$$P(E\mid \neg H,B) = \sum_{i=1}^{m} P(E\mid A_i,B)P(A_i\mid \neg H,B)$$

Conditional probablities are just weights. $\sum_{i=1}^{m} w_i = 1$, $w_i = P(A_i \mid \neg H, B)$

Improbable $\neq$ Impossible

04. Alternative explanations

- **Legend Theory** : Jesus could have never existed and the resurrection is a legend story.
- **Removal of the Body** : The body was taken from the tomb, possibly by the disciples or others.

05. Lims and Apps

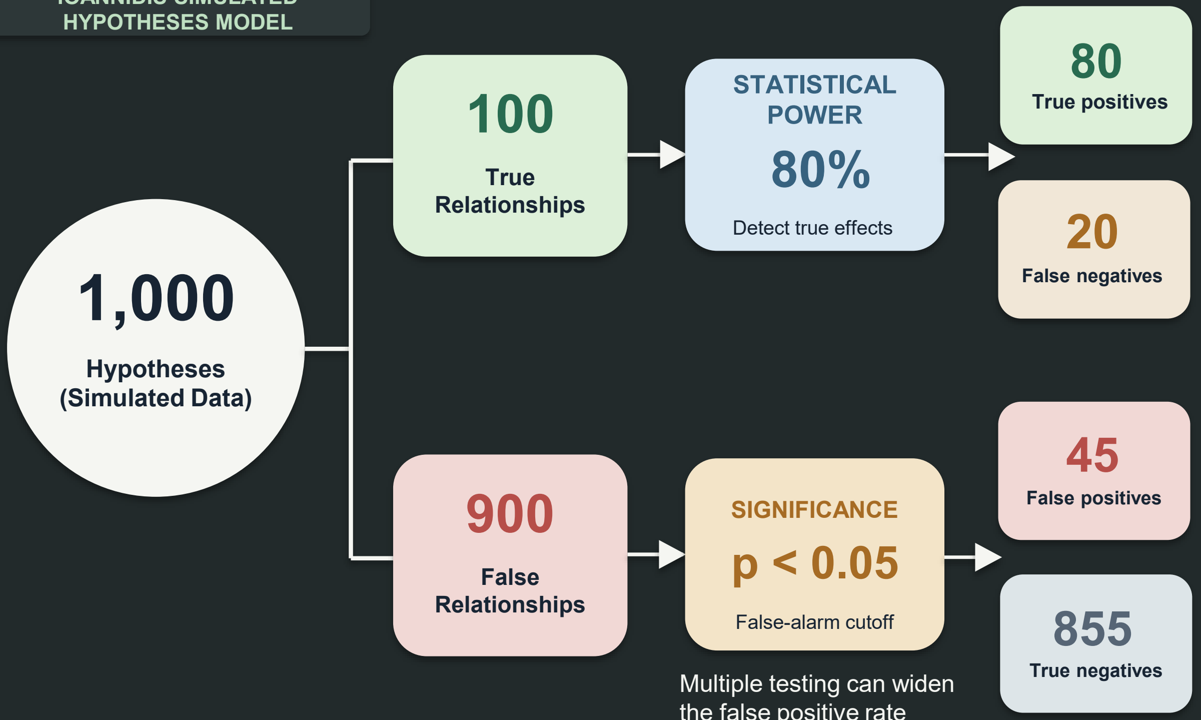
- **Dependence of evidences** : Evidences could be dependent.
- **Uncertain Probability** : Assignments Priors, likelihoods and alternative weights are hard to estimate.
- **Partition Assumption** .

- **Crimes -eg.** DNA testing.
- **Weather forecasting.**

Does the p < 0.05 rule reliably separate true findings from false ones?

Analysing the Pipeline of Falsehoods in published research findings: How often the scientific publication system gets things wrong and why?

IOANNIDIS SIMULATED HYPOTHESES MODEL



1,000
Hypotheses (Simulated Data)

100
True Relationships

STATISTICAL POWER
80%
Detect true effects

80
True positives

20
False negatives

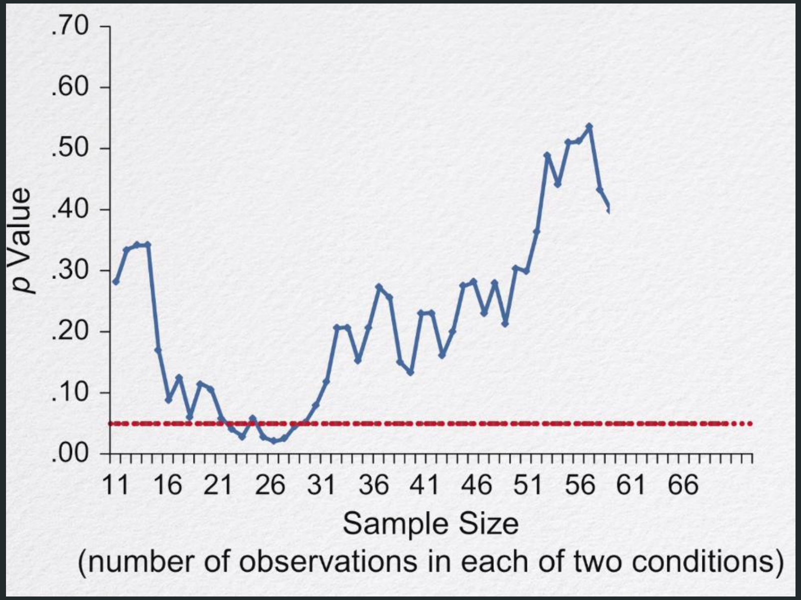
900
False Relationships

SIGNIFICANCE
$p < 0.05$
False-alarm cutoff

45
False positives

855
True negatives

Multiple testing can widen the false positive rate



P HACKING

METHODS

Simulates hypotheses to analyse published findings and checked effects of multiple testing, p hacking and reproducibility

KEY RESULT

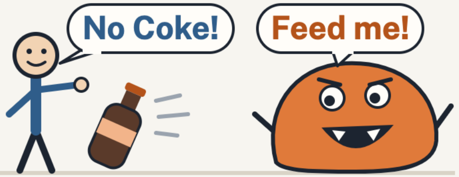
45 out of 145 published findings (31%) are false positives, highlighting publication bias

TAKEAWAY

Be critical of published results, especially those with low power or exploratory and reproducibility analysis

Media: “Is Most Published Research Wrong?”, Veritasium (YouTube), 11 Aug 2016
Timestamps: 0:00–1:08 (Bem 2011 Precognition Curtain Study Experiment), 2:00–3:14 (Ioannidis 2005 False Discovery Rate Model), 3:47 (Open Science Collaboration 2015 Reproducibility Project), 4:00 – (Begley & Ellis 2012 Landmark Cancer Study Replication) 4:45-5:10 (Bohannon 2015 chocolate diet study, p hacking)

Can cutting one Coke a day halve obesity?


No Coke!
Feed me!

1 · THE CLAIM

Cut 216 kcal a day, about one bottle of Coke, and UK obesity would halve.

Said by the Health Secretary, based on research by the charity Nesta. 00:50–01:40

→

2 · DATA USED

- Waist sizes now vs. the early 1990s 01:40
- Controlled feeding trials that measure intake exactly 03:45
- 2016 drug study: calories secretly removed, eating tracked 04:10

→

3 · ASSUMPTIONS

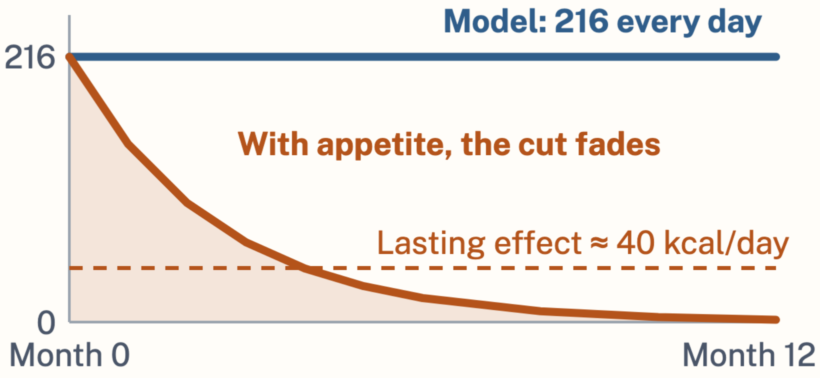
- 216 kcal is an average across everyone overweight 01:40
- Without the cut, people's weight would stay steady 03:20
- Lab results also hold in real life 03:45

→

4 · LEFT OUT

- **Appetite:** hunger rises as weight falls 04:10
- **Overweight people:** Streeting quoted it for obese people only 01:40
- **Accurate intake:** people misjudge what they eat 06:15

5 · THE DAILY CUT FADES OVER A YEAR

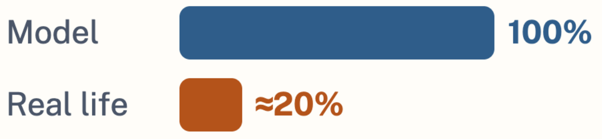


Each kg lost adds ~95 kcal/day of hunger 04:35

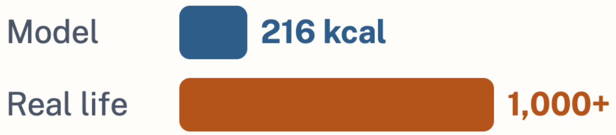
→

6 · HOW BIG IS THE EFFECT?

Weight actually lost 06:40



Day-1 cut needed to end at 216 07:05



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7 · IS IT CORRECT?

✓ **The maths: yes**

Nesta applied Hall's model correctly 03:20

⚠ **The promise: no**

Appetite cuts the real effect to ~20% 06:40

Nesta: still worth it at 1/10 of the effect 08:20

Vikas Jha · DA26D401
MA 5104: Statistical Inference and Learning · Instructor: Dr. Rakhi Singh

Source: "Can drinking one less bottle of coke a day halve obesity?", *More or Less*, BBC Radio 4, 9 July 2025. Curve in 5 is illustrative.

Can We Predict Rain?

A probabilistic approach using statistical learning

Statistical Inference and Learning | Parth & Prakash

CONTEXT

Weather data can help us estimate the likelihood of rainfall— not just predict “ rain” or “ no rain.”

HOW IS THE PROBABILITY OF RAINFALL ESTIMATED?

1 WEATHER OBSERVATIONS

Temperature • Humidity
Pressure • Wind • Cloud cover

2 STATISTICAL MODEL

Logistic regression
or ensemble forecasting

3 PREDICTED PROBABILITY

40%

Illustrative estimate

METHODS

Conditional probability and logistic regression; ensemble forecasts represent model uncertainty.

KEY RESULT

A probability communicates uncertainty. A 40% forecast is not a guarantee of rain.

TAKEAWAY

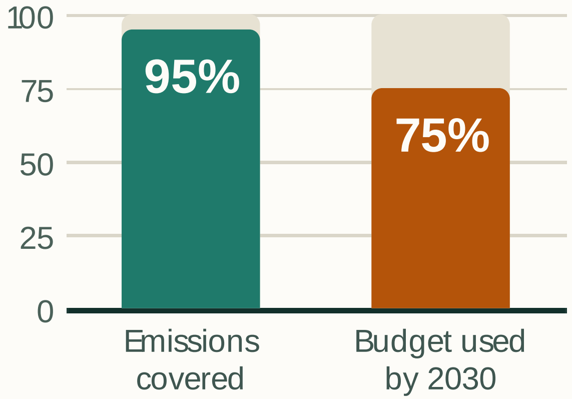
Calibrated forecasts can support farming, travel planning and flood preparedness.

Source: Weather Geeks Podcast, Episode 353 – "Why 40% Chance of Rain Confuses Everyone " (June 4, 2025).

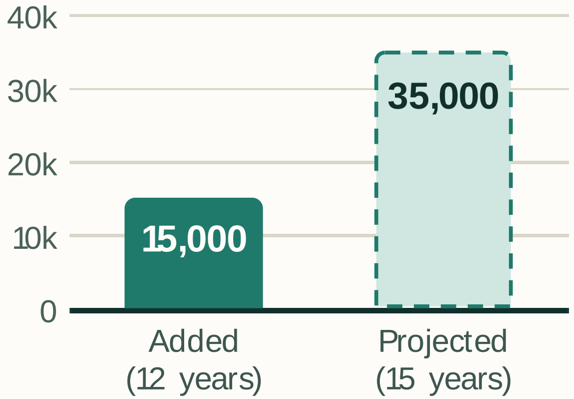
Climate Momentum Is Real, But Pledges Still Leave an Ambition Gap

A Paris COP21 briefing counts support and jobs, then measures what pledges miss.

Reported shares (%)



Minnesota clean-energy jobs



More numbers cited

Value	Measures
400,000	Climate marchers
180+	Nations pledged
2	Coal plants closing
~40 / yr	Deaths, one coal plant

No figure has a margin of error.

Methods: Descriptive counts, proportions, projection; no sample size or margin of error.

Key result: Pledges cover about 95% of emissions, yet leave an ambition gap.

Takeaway: Without uncertainty ranges, an encouraging pattern cannot support a causal claim.

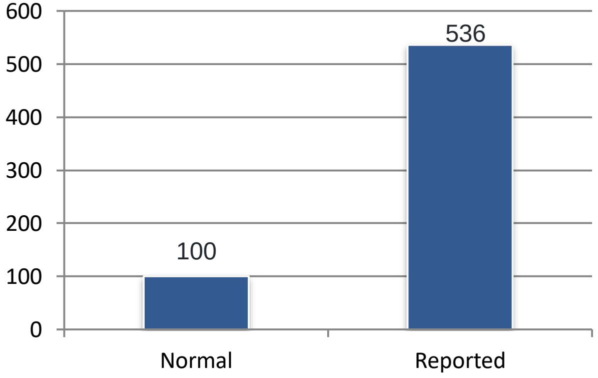
Source: [COP 21 USA and China Commitments to New Climate Change Regulations] · https://youtu.be/0R1SnJLOXf0?si=JB6MFC9A21qJ8Su9

READING THE FLOOD NUMBERS

How statistics turn rainfall data into Assam’s flood warnings

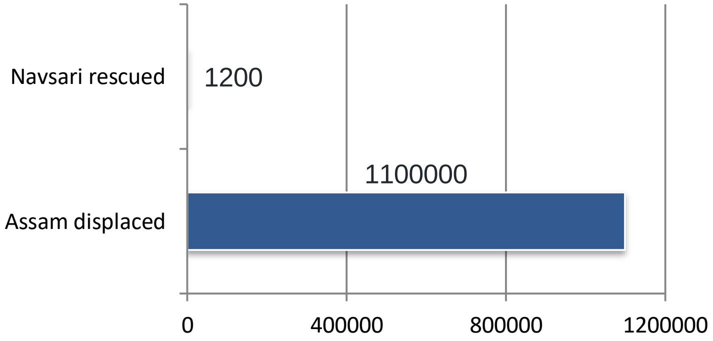
THE QUESTION : When does extreme rainfall become a flood warning?

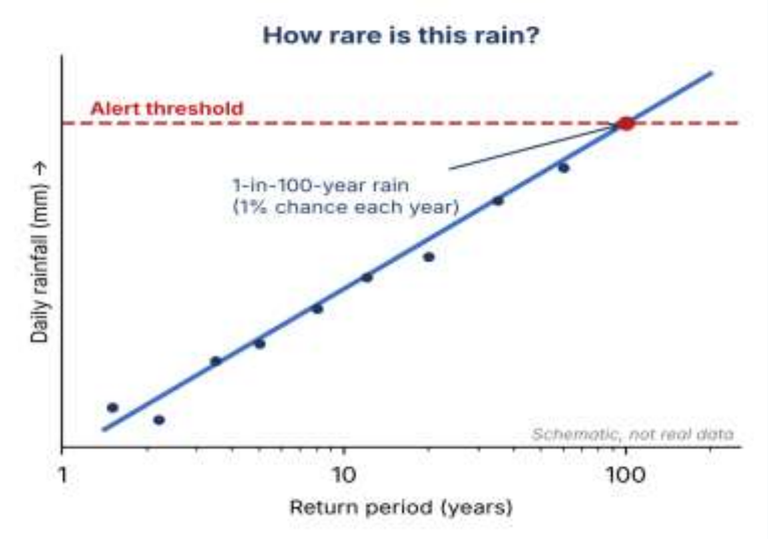
RAINFALL ANOMALY · CHARAIDEO



Rainfall index (normal = 100)

PEOPLE AFFECTED · TWO LOCATIONS





68 deaths
Assam [00:11–00:20]

>1,000 homes
Destroyed [00:11–00:20]

>4,000 km
Embankments [03:34–04:12]

Early warnings
Odisha example [08:22–09:15]

Methods- Rainfall and river-level records, frequency analysis, return periods and threshold-based alerts.

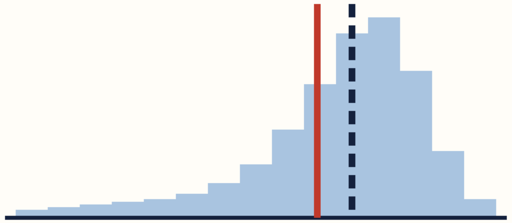
Key result- Extreme rainfall is increasing, but embankments, mining and planning decide who floods.

Takeaway- Warnings work only when statistics meet local data, infrastructure and preparedness.

Statistical Thinking in Science

1 Basic Statistics

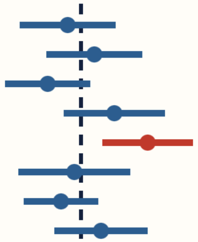
- Mean, median, mode = the center
- Spread: how much values vary
- Sample must be random and representative



Red mean, navy median

2 *Confidence Intervals*

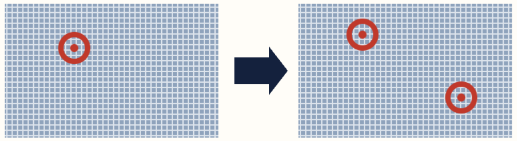
- A range, not a single number
- 95%: the method hits the truth 95% of the time
- Narrower range = more precise



Each line = one study

3 *Relative vs. Absolute Risk*

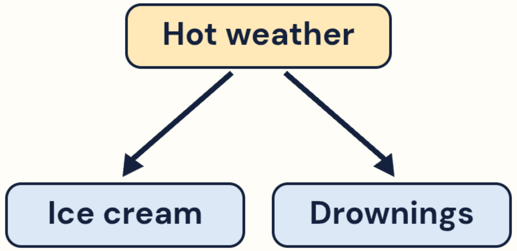
- Relative: "doubles the risk"
- Absolute: 1 in 1,000 to 2 in 1,000
- Ask: relative to what baseline?



1 vs 2 per 1,000: one extra

4 *Correlation vs. Causation*

- Moving together is not causing
- Confounders, reverse cause, chance
- Randomized trials get closer



Hot weather

Ice cream

Drownings

Heat drives both

BIG IDEA

Source: CrashCourse, Scientific Thinking #2 (2026). Diagrams are original illustrations.

Can Randomness Win? Game Theory Behind Football Penalty Kicks.

On the penalty spot, the smartest move may be the one nobody predicts.

CONDITIONAL SCORING PROBABILITY

2,717 penalties (1995–2004) [S]

		GOALKEEPER DIVES	
		LEFT	RIGHT
KICKER SHOOTS	LEFT	60% DIVE MATCHES	95% WRONG-FOOTED
	RIGHT	90% WRONG-FOOTED	70% DIVE MATCHES

MINIMAX MIX

36%

KICKER SHOOTS LEFT

45%

KEEPER DIVES LEFT

WHAT THE EVIDENCE SAYS

- **METHODS**

  Video-code penalty choices, compare outcomes, derive mixed-strategy probabilities. [1][2]

- **KEY RESULT**

  Experts randomize effectively, Chelsea revealed a costly directional pattern. [2][3]

- **TAKEAWAY**

  Strategic randomness is weighted, empirical, and never static. [3][4]

SOURCE NPR Planet Money - “How to win a penalty shootout (with game theory)” - 3 July 2026

WATCH ↗

[1] 03:27–04:20, [2] 05:52–06:26; 08:23–08:41, [3] 09:19–13:31, [4] 15:55–16:32; 17:21–18:08

[S] Supplementary data, not from the audio: Palacios-Huerta & Volij (2005), “Natural Minimax Players,” page 9.